

GAURAV DULAL

ML ENGINEER · RESEARCHER

CONTACT

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TECHNICAL SKILLS

LANGUAGES

Python · Java · C · JavaScript

ML / DL

Scikit-learn · TensorFlow · XGBoost

DATA

NumPy · Pandas · Matplotlib · Seaborn

NLP

NLTK · TF-IDF · LSTM · Transformers

DATABASE

PostgreSQL · SQL

DEV TOOLS

Git · GitHub · VS Code · Streamlit

WEB

HTML · CSS · React basics

STRENGTHS

- Independent learner
- Research mindset
- Daily builder (158+ streak)
- End-to-end delivery
- Technical writing

LANGUAGES

English	IELTS 7.0
Nepali	Native
German	Learning (A1)

Gaurav Dulal

BHons Computing Student | Machine Learning Engineer (in progress) | AI Researcher

PROFILE SUMMARY

Self-driven computing student and ML practitioner from Belbari, Nepal, with a 158-day unbroken GitHub commit streak and 1,900+ contributions. Completed a full classical ML stack independently — from NumPy through XGBoost and Ensemble methods — alongside formal BHons Computing studies at Itahari International College (London Metropolitan University). Undergraduate researcher, deployed application builder, and aspiring MSc ML candidate targeting the University of Tübingen.

EDUCATION

BHons Computing — Itahari International College

2025 – 2028 | Morang, Nepal

Affiliated with London Metropolitan University, UK

- Year 1 completed with distinction — 100/100 on Programming exam; A grade in Information Systems
- Modules: Programming (Java/Python), Fundamentals of Computing, Computer Hardware & Software Architectures, Logic & Problem Solving
- Summer Enrichment: Applied AI and Data Analysis — IIC internal programme

MACHINE LEARNING EXPERTISE

- Data foundations: NumPy, Pandas, Matplotlib, Seaborn, Cufflinks — EDA, cleaning, feature engineering
- Classical ML (Scikit-learn): Linear & Logistic Regression, KNN, Naive Bayes, Decision Trees, SVM, Random Forest
- Advanced ML: XGBoost, AdaBoost, Gradient Boosting, GridSearchCV, cross-validation, ensemble methods
- Unsupervised learning: KMeans, DBSCAN, PCA dimensionality reduction
- Statistics: hypothesis testing, probability distributions, Bayes theorem, confidence intervals
- SQL: PostgreSQL — CRUD, joins, aggregations, window functions, views
- NLP: in progress — TF-IDF, LSTM, Transformers
- Deep Learning: in progress — ANN, CNN, RNN (TensorFlow/Keras)
- Deployment: Streamlit web applications; model evaluation with F1, precision, recall, confusion matrix

PROJECTS

OOP Purity Analyzer

Flask · GitHub API · Plotly · Scikit-learn

- Research tool implementing a 7-category 24-sub-criterion OOP purity rubric. Scrapes GitHub repos, scores code quality 0-100, and presents results in an interactive Plotly dashboard with CSV export. Underpins undergraduate research paper.

WhatsApp Chat Analyzer

Streamlit · Pandas · Matplotlib · Seaborn · WordCloud

- End-to-end Streamlit application that parses WhatsApp .txt exports, performs statistical analysis, and visualises message frequency, word clouds, most common words, and user activity patterns.

Titanic ML — Kaggle

Scikit-learn · Pandas · XGBoost

- First Kaggle competition submission achieving 78% accuracy on the classic binary classification challenge using feature engineering and ensemble methods.

ML-Foundations Repository

Python · Jupyter · NumPy · Pandas · Scikit-learn

- Public GitHub repository documenting the complete ML learning journey — NumPy through Deep Learning — with structured notebooks, real datasets, and daily commits. 1,900+ contributions.

RESEARCH

A Quantitative Framework for Evaluating OOP Purity in Modern Programming Languages

- Undergraduate research paper — team of 4, targeting journal publication
- Proposes a scoring rubric across 8 measurable OOP dimensions (encapsulation, inheritance, polymorphism, abstraction, and more)
- Languages evaluated: Java, Python, Ruby, Kotlin, C++, JavaScript
- Implemented as the OOP Purity Analyzer tool — automated validation of the framework

ACHIEVEMENTS & STATS

- 158-day consecutive GitHub commit streak (Mar 2026 – Aug 2026) — unbroken through exams, vivas, and holidays
- 1,900+ GitHub contributions — Jupyter Notebook now #1 language at 45% of codebase
- 100/100 on Year 1 Programming exam; A grade in Information Systems; top results across all modules
- IELTS 7.0 — English proficiency meeting University of Tübingen MSc ML requirements
- LeetCode: 57+ problems solved including Medium difficulty — dedicated DSA practice repository
- Hackathon participant — ClashATHon: StrawHats-RouteRight collaborative JavaScript project